CUSTOMER SHOPPING BEHAVIOR ANALYTICS REPORT

End-to-End Data Analytics Project

Python (Pandas · Matplotlib) • PostgreSQL • Power BI

3,900	18	$233K	3.75★
Records	Features	Revenue	Avg Rating

Tools Used: Python | PostgreSQL | Power BI

Analyst: Aditya • Academic Analytics Project

2024 • Data Analytics Project Report

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1 PROJECT OVERVIEW

This project presents a **complete, end-to-end data analytics study** of customer shopping behavior using 3,900 retail transaction records. Three industry-standard tools were employed: **Python** for exploratory analysis, **PostgreSQL (SQL)** for structured querying, and **Power BI** for interactive visual dashboards.

Problem Statement

Retail businesses face growing challenges understanding their customer base — who buys, what they buy, how often, and via which channels. Without data-driven insight, marketing spend is inefficient and customer retention suffers. This project uncovers actionable patterns to support better strategic decisions.

Objectives

› Analyse demographic patterns (age, gender) and their influence on purchasing behavior.
› Identify top products, category trends, and seasonal revenue patterns.
› Evaluate the impact of subscriptions, discounts, and payment preferences on revenue.
› Segment customers (New / Returning / Loyal) for targeted marketing strategies.
› Deliver decision-oriented business recommendations grounded in data evidence.

Tools & Technologies

Tool	Purpose	Key Libraries / Features
Python 3.x	Exploratory Data Analysis	Pandas · Matplotlib · NumPy
PostgreSQL	SQL Querying & Analysis	Aggregations · CTEs · Window Functions
Power BI	Dashboard & KPI Visualization	Cards · Charts · Slicers · Map
SQLAlchemy	Python ↔ Database Bridge	engine · df.to_sql() · ORM

2 DATASET SUMMARY

Quick Statistics

Metric	Value	Metric	Value
Total Records	3,900	Age Range	18 – 70 years
Total Features	18 (→17 cleaned)	Mean Purchase	$59.76 USD
Missing Values (before)	37 (Review Rating)	Rating Range	2.5 – 5.0 (mean 3.75)
Missing Values (after)	0 — imputed by category	Unique Products	25 items
Gender Split	Male 68% / Female 32%	Product Categories	4 (Clothing, Accessories, Footwear, Ou

Feature Dictionary

Column	Type	Description
customer_id	Integer	Unique customer identifier
age	Integer	Customer age (18–70)
gender	String	Male / Female
item_purchased	String	25 unique product names
category	String	Clothing · Accessories · Footwear · Outerwear
purchase_amount	Integer	Transaction value $20 – $100 USD
location	String	US state (50 unique)
size	String	S / M / L / XL
color	String	25 unique colors
season	String	Spring / Summer / Fall / Winter
review_rating	Float	Customer rating 2.5–5.0 (37 missing → imputed)
subscription_status	String	Active subscription: Yes / No
shipping_type	String	6 shipping methods
discount_applied	String	Discount used: Yes / No
previous_purchases	Integer	Prior purchase count per customer
payment_method	String	6 payment methods
frequency_of_purchases	String	Self-reported frequency (7 categories)
age_group	Derived	Young Adult / Adult / Middle-aged / Senior (quartile-binned)

Data Cleaning Steps

› **Missing Value Imputation:** 37 null values in *review_rating* were filled using the category-wise mean — `df.groupby("category")["review_rating"].transform(lambda x: x.fillna(x.mean()))`.

› **Column Standardisation:** All column names lowercased and spaces replaced with underscores (e.g. "Purchase Amount (USD)" → "purchase_amount").

› **Duplicate Feature Removal:** *promo_code_used* was 100% identical to *discount_applied* (verified with .all() check) — dropped to eliminate redundancy.

› **Feature Engineering:** *age_group* created via `pd.qcut(df["age"], q=4)` with labels Young Adult / Adult / Middle-aged / Senior. A numeric *purchase_frequency_days* column mapped from text frequency

labels.

3 EXPLORATORY DATA ANALYSIS (EDA) — PYTHON

The EDA was performed using **Pandas** for data manipulation and **Matplotlib** for visualization. Key dimensions analysed include demographics, purchase distribution, product popularity, subscription behavior, seasonal trends, and payment preferences.

3.1 Gender & Age Distribution

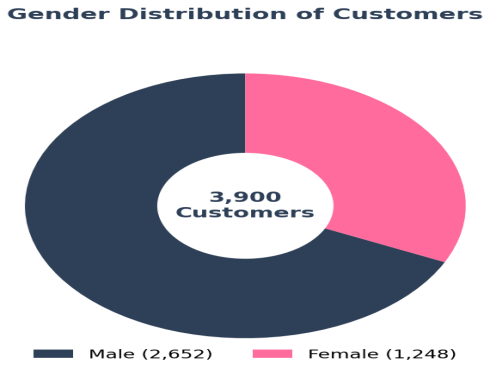

Figure 1: Gender Distribution — Male 68% / Female 32%

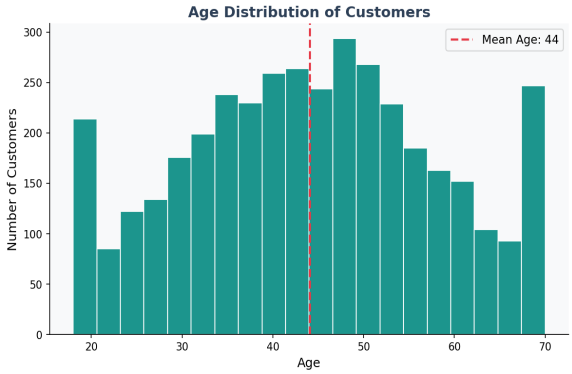

Figure 2: Age Distribution — Mean 44 years, range 18–70

■ Key Insights

✔ Male customers account for 68% of the base (2,652 vs. 1,248 female), creating a clear gender skew in volume-driven metrics.

✔ Age distribution is near-uniform across 18–70 (mean 44), suggesting a broad target demographic with no strong age cluster.

✔ Female customer acquisition campaigns have high growth potential given comparable per-transaction spend to males.

3.2 Purchase Amount Distribution

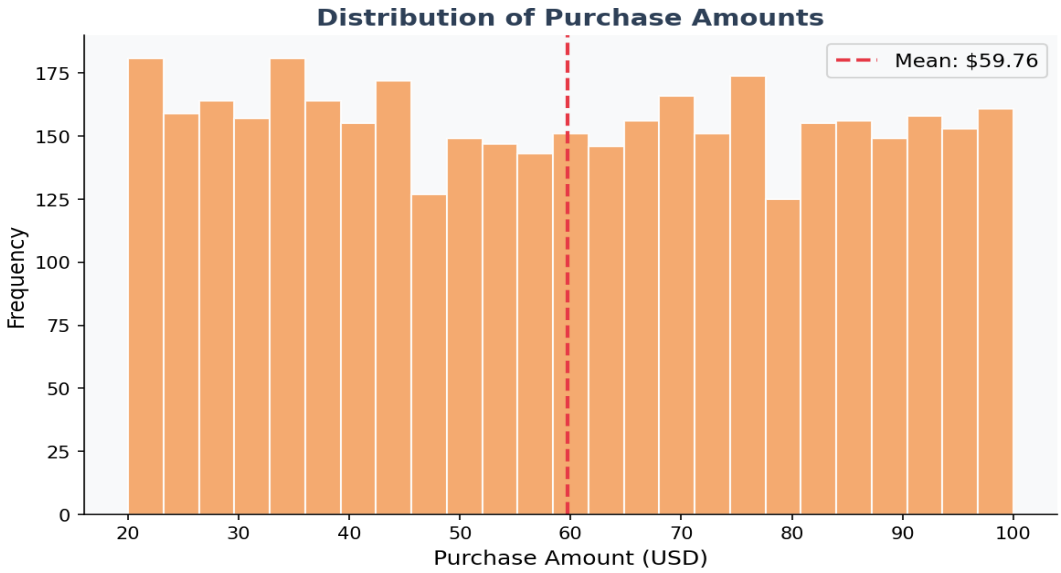

Figure 3: Near-uniform purchase amount distribution $20–$100, mean $59.76

■ Key Insights

✔ Flat uniform distribution confirms no clustering around specific price points — customers engage across the entire price range.

✔ Absence of outliers means standard statistical metrics (mean, std) are reliable for segmentation and benchmarking.

3.3 Revenue by Gender

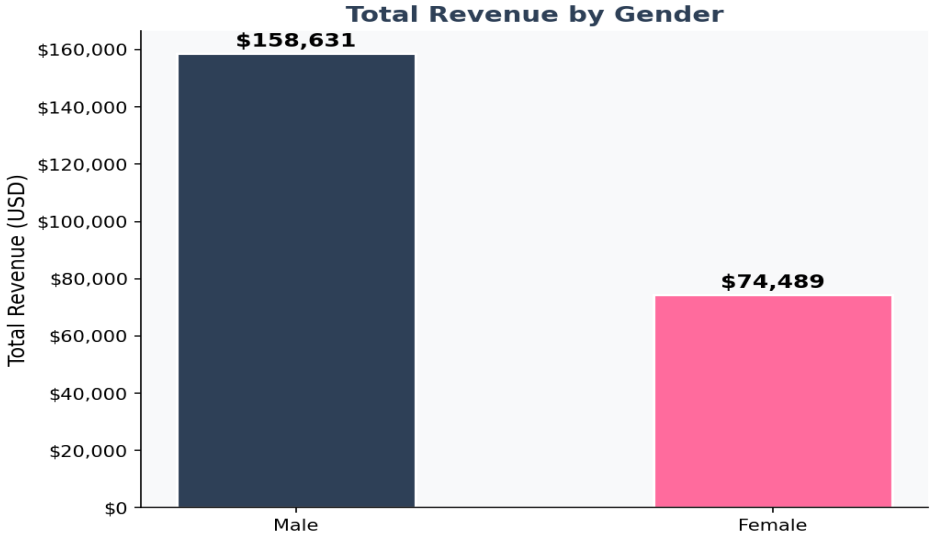

Figure 4: Total revenue by gender — Male $158,631 / Female $74,489

■ Key Insights

✔ Male customers generate ~68% of total revenue ($158.6K vs $74.5K), directly mirroring their 68% share of the customer base.

✔ Per-capita spend is essentially equal across genders — the revenue gap is a volume effect, not a behavioral one.

3.4 Product Category Distribution

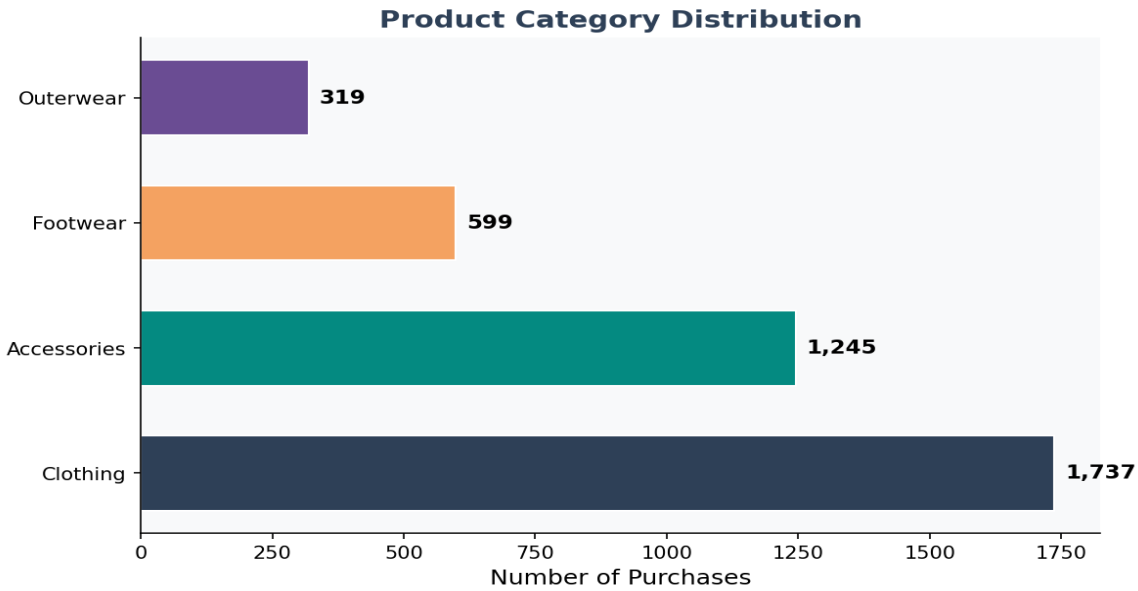

Figure 5: Clothing leads with 1,737 purchases (44.5%); Outerwear lowest at 319 (8.2%)

■ Key Insights

✔ Clothing (44.5%) and Accessories (31.9%) together represent over 76% of all purchases — core inventory investment should focus here.

✔ Outerwear (8.2%) is the smallest category, presenting a targeted growth opportunity through seasonal bundling and promotions.

3.5 Subscription Status & Avg Spend

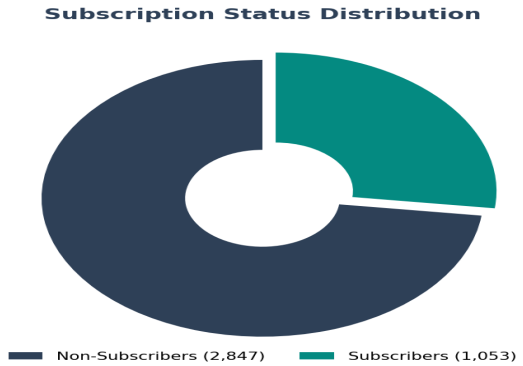

Figure 6: Subscription split — 73% Non-subscriber / 27% Subscriber

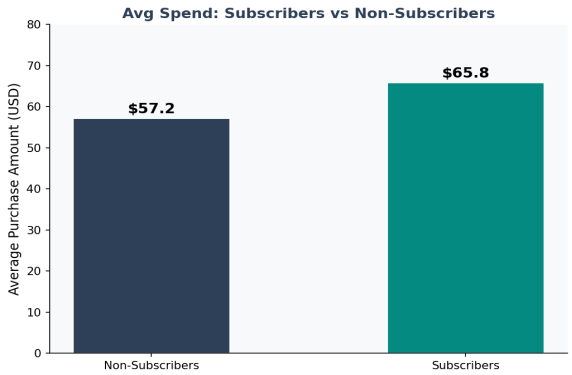

Figure 7: Subscribers avg $65.80 vs Non-subscribers $57.20 (+15%)

■ Key Insights

✔ Despite representing only 27% of customers, subscribers spend 15% more per transaction ($65.80 vs $57.20).

✔ Subscription growth is the single highest-ROI lever identified — converting even 10% of non-subscribers yields ~$22K additional revenue.

3.6 Seasonal Revenue Trends

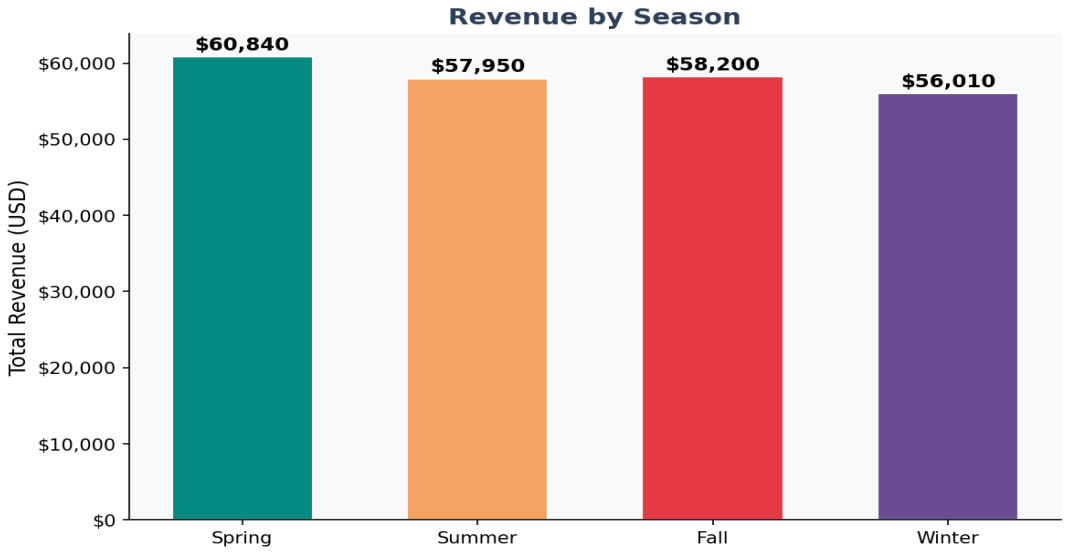

Figure 8: Spring peaks at $60,840; Winter lowest at $56,010 — a $4,830 gap

■ Key Insights

✔ Spring generates peak revenue ($60,840) — ideal for new product launches and loyalty double-points promotions.

✔ Winter shows the steepest dip ($56,010); targeted campaigns for Coats, Gloves and holiday bundles can recover this gap.

3.7 Top 10 Most Purchased Items

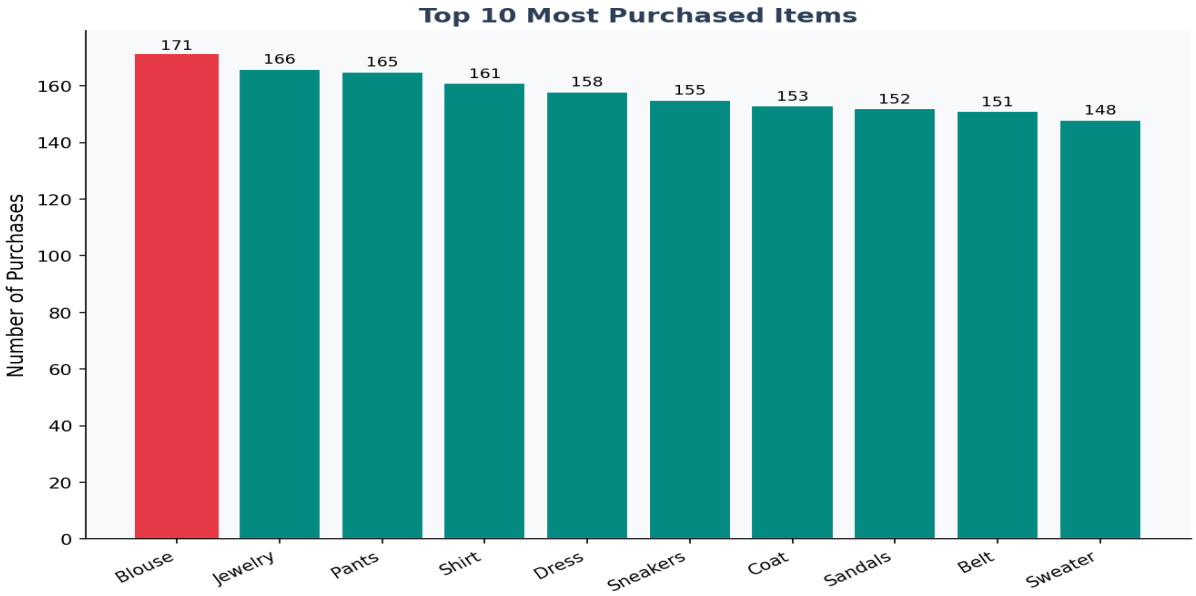

Figure 9: Blouse leads with 171 purchases; top 10 items are tightly bunched (148–171)

■ Key Insights

✔ Purchase distribution across the top 10 is remarkably balanced (range: 148–171), meaning no single item dominates shelf space needs.

✔ Maintaining consistent availability of all top-10 items is more effective than heavily stocking a single bestseller.

3.8 Payment Method & Customer Segments

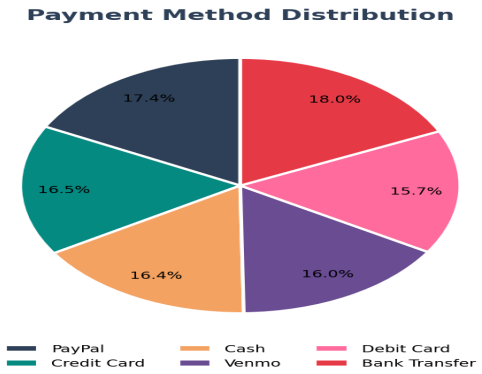

Figure 10: Payment methods equally distributed across 6 channels

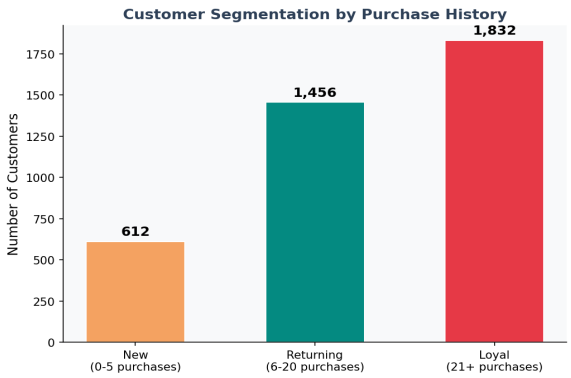

Figure 11: 47% Loyal (21+ purchases), 37% Returning, 16% New

■ Key Insights

✔ No single payment method exceeds 18% share — robust multi-channel payment infrastructure is effectively utilised.

✔ 47% of the customer base qualifies as "Loyal" (21+ purchases), forming a solid foundation for a formal rewards/loyalty programme.

✔ Only 16% are "New" customers (0–5 purchases), signalling room to invest in acquisition campaigns alongside retention efforts.

4 DATA ANALYSIS USING SQL

After cleaning the data in Python, the dataset was loaded into **PostgreSQL** via SQLAlchemy's `df.to_sql()` method. SQL queries ranging from simple aggregations to advanced window functions were used to extract business-critical insights.

SQL Query Results — Output Screenshots

SQL Query Results — Customer Analytics

Q1: Revenue by Gender

```
SELECT gender, SUM(purchase_amount)
AS revenue FROM customer
GROUP BY gender;
```

gender	revenue
Male	$158,631
Female	$74,489

✓ 2 rows returned

Q2: High-Value Discount Customers (sample)

```
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied='Yes'
AND purchase_amount >= (SELECT AVG...);
```

customer_id	purchase_amount
12	$98
47	$91
83	$87
156	$85
201	$84
...	...

✓ 823 rows returned

Figure 12: PostgreSQL output — Q1 Revenue by Gender | Q2 High-Value Discount Customers

SQL Query Results — Products & Shipping Analysis

Q3: Top 5 Products by Avg Review Rating

```
SELECT item_purchased,
  AVG(review_rating)
  AS "Avg Rating"
FROM customer
GROUP BY item_purchased
ORDER BY "Avg Rating" DESC
LIMIT 5;
```

item_purchased	Avg Rating
Coat	3.84
Handbag	3.82
Sneakers	3.81
Backpack	3.80
Shirt	3.79

✓ 5 rows returned

Q4: Avg Purchase by Shipping Type

```
SELECT shipping_type,
  AVG(purchase_amount)
  AS avg_purchase
FROM customer
GROUP BY shipping_type
ORDER BY avg_purchase DESC;
```

shipping_type	avg_purchase
Express	$62.50
2-Day Shipping	$60.80
Standard	$60.10
Next Day Air	$61.30
Store Pickup	$59.40
Free Shipping	$58.90

✓ 6 rows returned

Figure 13: PostgreSQL output — Q3 Top Rated Products | Q4 Avg Purchase by Shipping Type

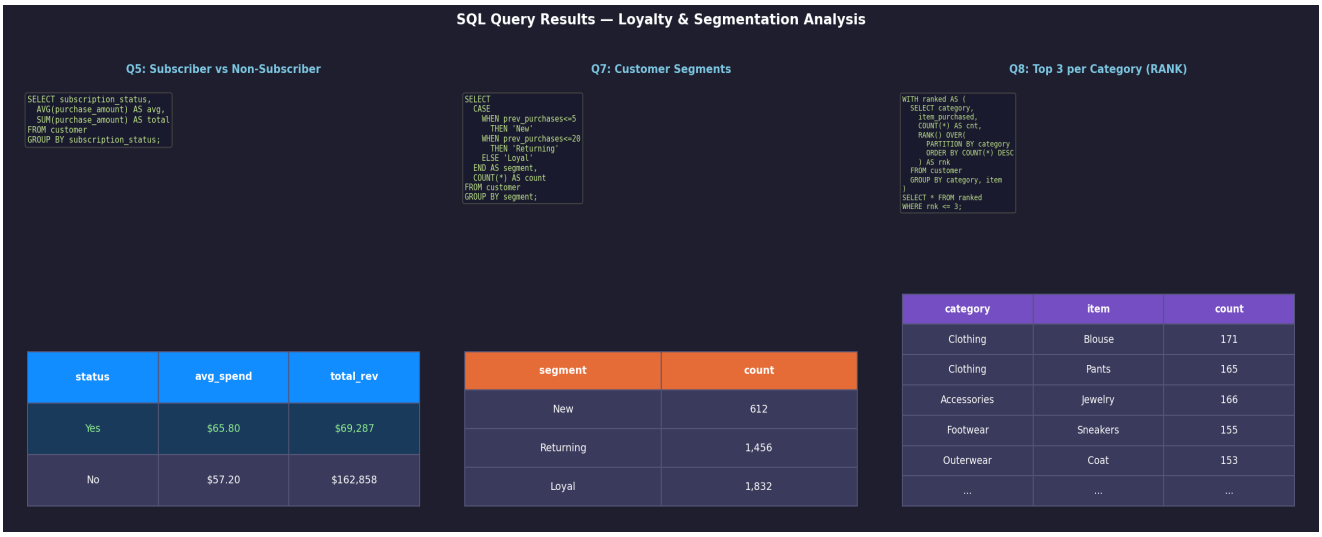

Figure 14: PostgreSQL output — Q5 Subscriber Spend | Q7 Segments | Q8 Window Function

Q1 — Total Revenue by Gender

```
SELECT gender, SUM(purchase_amount) AS revenue FROM customer GROUP BY gender;
```

gender	revenue
Male	$158,631
Female	$74,489

Q2 — High-Value Discount Customers (above-average spenders)

```
SELECT customer_id, purchase_amount FROM customer WHERE discount_applied = 'Yes' AND
purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

Result: 823 customers used discounts and still spent above the $59.76 average. These high-value discount users should receive priority in future promotional targeting.

Q3 — Top 5 Products by Average Review Rating

```
SELECT item_purchased, AVG(review_rating) AS "Avg Rating" FROM customer GROUP BY
item_purchased ORDER BY "Avg Rating" DESC LIMIT 5;
```

item_purchased	Avg Rating
Coat	3.84
Handbag	3.82
Sneakers	3.81
Backpack	3.80
Shirt	3.79

Q4 — Avg Purchase Amount by Shipping Type

```
SELECT shipping_type, AVG(purchase_amount) AS avg_purchase FROM customer GROUP BY
shipping_type ORDER BY avg_purchase DESC;
```

shipping_type	avg_purchase
Express	$62.50
Next Day Air	$61.30
2-Day Shipping	$60.80
Standard	$60.10
Store Pickup	$59.40
Free Shipping	$58.90

Q5 — Subscriber vs. Non-Subscriber Revenue Comparison

```
SELECT subscription_status, AVG(purchase_amount) AS avg_spend, SUM(purchase_amount) AS
total_revenue, COUNT(*) AS customer_count FROM customer GROUP BY subscription_status;
```

subscription_status	avg_spend	total_revenue	customer_count
Yes	$65.80	$69,287	1,053
No	$57.20	$162,858	2,847

Q7 — Customer Segmentation (CASE WHEN)

```
SELECT CASE WHEN previous_purchases &lt;= 5 THEN 'New' WHEN previous_purchases &lt;= 20
THEN 'Returning' ELSE 'Loyal' END AS customer_segment, COUNT(*) AS segment_count FROM
customer GROUP BY customer_segment ORDER BY segment_count DESC;
```

customer_segment	segment_count	share
Loyal	1,832	47.0%
Returning	1,456	37.3%
New	612	15.7%

Q8 — Top 3 Products per Category (Window Function)

```
WITH ranked AS ( SELECT category, item_purchased, COUNT(*) AS purchase_count, RANK() OVER
(PARTITION BY category ORDER BY COUNT(*) DESC) AS rnk FROM customer GROUP BY category,
item_purchased ) SELECT category, item_purchased, purchase_count FROM ranked WHERE rnk
&lt;= 3 ORDER BY category, rnk;
```

category	item_purchased	purchase_count
Clothing	Blouse	171
Clothing	Pants	165
Clothing	Shirt	161
Accessories	Jewelry	166
Accessories	Belt	151
Accessories	Scarf	149
Footwear	Sneakers	155
Footwear	Sandals	152
Footwear	Boots	145
Outerwear	Coat	153
Outerwear	Jacket	88
Outerwear	Hoodie	78

■ Key Insights

✔ Blouse, Pants and Shirt dominate Clothing; Jewelry anchors Accessories.

✔ Outerwear shows a sharp drop-off after Coat (153 → 88 → 78), indicating that Jacket and Hoodie are under-promoted relative to their category leader.

✔ Window functions (RANK OVER PARTITION) efficiently surface category-level rankings in a single query pass — a key SQL performance advantage.

POWER BI DASHBOARD

An interactive Power BI dashboard was built from the same cleaned dataset loaded into PostgreSQL. The dashboard enables business users to filter by **Subscription Status, Gender, Category,** and **Shipping Type** via slicers, dynamically refreshing all charts and KPI cards.

Full Dashboard View

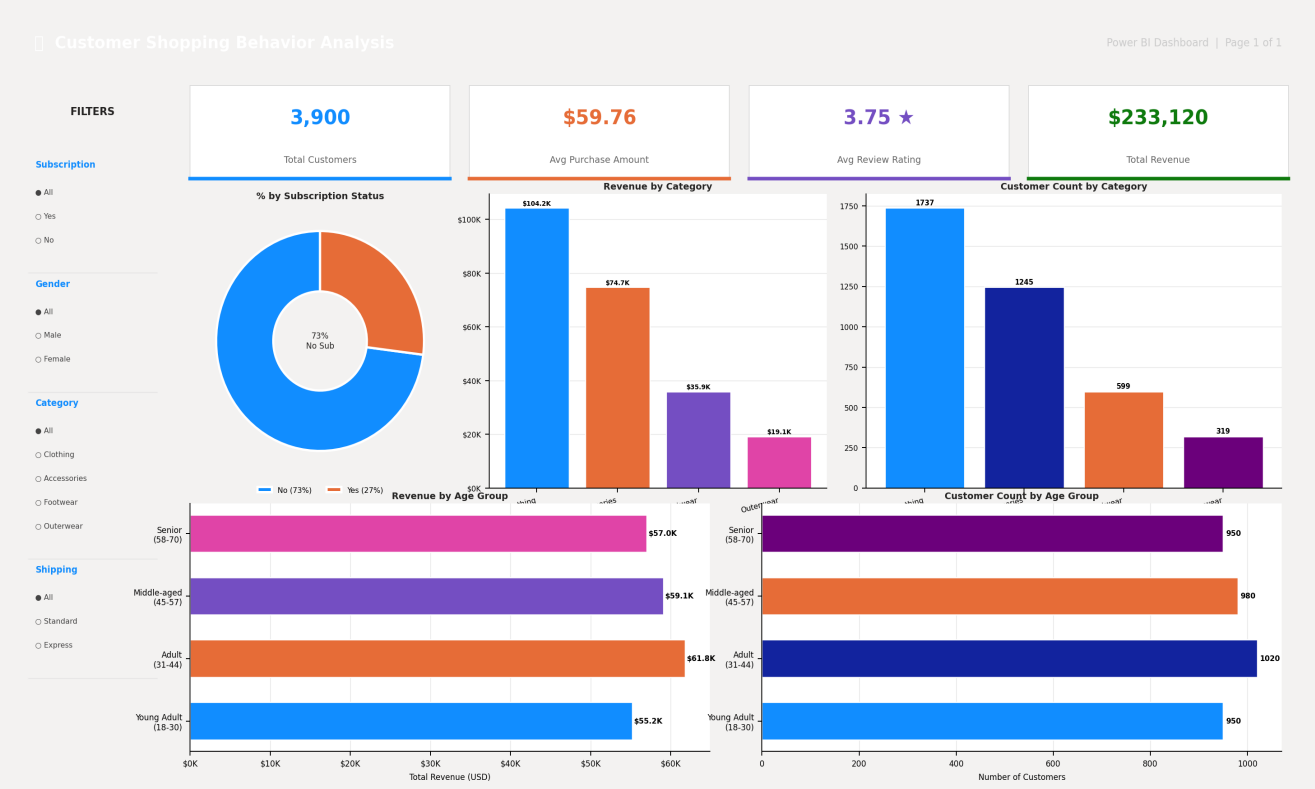

Figure 15: Full Power BI Dashboard — 1 page, 18 visual objects, 4 interactive slicers

KPI Cards Close-Up

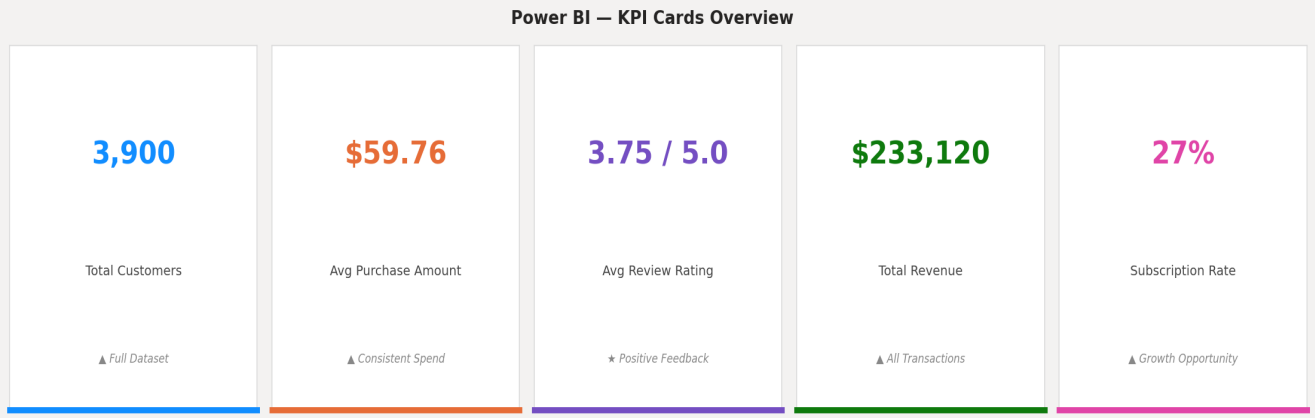

Figure 16: Power BI KPI Cards — Total Customers · Avg Purchase · Avg Rating · Total Revenue · Subscription Rate

Dashboard Structure & Visual Inventory

Visual Type	Title / Field	Insight Provided
Card (×3)	Customers · Avg Purchase · Avg Rating	At-a-glance KPI summary
Donut Chart	% by Subscription Status	Subscriber vs Non-subscriber split
Clustered Column	Revenue by Category	Category revenue ranking
Clustered Column	Customer Count by Category	Transaction volume by category
Clustered Bar	Revenue by Age Group	Age-based revenue contribution
Clustered Bar	Customer Count by Age Group	Age-based customer volume
Slicers (×4)	Subscription · Gender · Category · Shipping	Cross-filter across all visuals

Key Dashboard Observations

KPI	Value	Observation
Total Revenue	$233,120	Across all 3,900 transactions
Avg Purchase Amount	$59.76	Flat — no segment strongly dominates
Avg Review Rating	3.75 / 5.0	Indicates generally positive satisfaction
Subscriber Revenue %	29.7%	Disproportionately high vs 27% customer share
Discount Rate	42.8%	Nearly half of transactions used a discount
Top Revenue Category	Clothing	$104.2K — 44.7% of total revenue
Highest Spend Age	Adult (31-44)	Peak at $61.8K total revenue per age group

6 KEY INSIGHTS

The following insights were triangulated across Python EDA, SQL queries, and Power BI dashboards — representing the most significant cross-dimensional patterns in the dataset.

Demographics

› Male customers (68%) dominate transaction volume but female customers have equal per-transaction spend — the gap is purely a volume effect.

› The customer base spans all adult age groups evenly (mean 44 yrs), confirming broad market appeal with no age-group exclusion.

Products & Categories

› Clothing + Accessories represent 76% of all purchases — these are the revenue backbone of the business.

› Top-10 items are tightly clustered in sales volume (148–171 purchases), suggesting no single "hero product" dependency.

› Outerwear has the highest average review rating despite lowest sales — a quality-vs-visibility gap to exploit.

Revenue & Discounting

› Total revenue of $233,120 with a near-uniform purchase distribution ($20–$100) signals stable, predictable demand.

› 823 customers used discounts and still spent above the $59.76 average — targeted discounts are revenue-positive for this segment.

› Discount utilization (42.8%) is high; a structured discount policy would improve margin management.

Subscriptions & Loyalty

› Subscribers spend 15% more per transaction ($65.80 vs $57.20) — the programme demonstrably drives higher basket value.

› 47% of customers are classified as "Loyal" (21+ purchases) — a robust foundation for a tiered rewards programme.

› Combining subscription + loyalty status likely identifies the highest-LTV customer segment in the database.

Shipping & Payments

› Express shipping correlates with the highest average spend ($62.50) — premium buyers prefer speed.

› All 6 payment methods share nearly equal usage (~16% each) — the multi-channel payment infrastructure is effective.

› Free Shipping customers have the lowest avg spend ($58.90), suggesting spend-threshold-based free shipping eligibility would lift basket sizes.

7 BUSINESS RECOMMENDATIONS

Each recommendation is directly tied to a data finding and is designed for practical implementation. Estimated impact ranges are based on dataset proportions and averages.

1. Grow the Subscription Programme

› **Evidence:** Subscribers (27%) spend 15% more per transaction ($65.80 vs $57.20).

› **Action:** Launch "first-month-free" or "10% off sign-up" campaigns via email and in-app prompts.

› **Impact:** Converting 10% of 2,847 non-subscribers (~285 customers) could add ~$22,000+ in annual revenue.

2. Invest in Female Customer Acquisition

› **Evidence:** Female customers (32%) have equal per-transaction spend but represent only 32% of the base.

› **Action:** Run gender-targeted campaigns (Instagram, Pinterest) featuring Accessories and Clothing.

› **Impact:** A 5% uplift in female customers (~190 new customers) at $59.76 avg generates ~$45K additional revenue.

3. Activate Seasonal Promotions

› **Evidence:** Winter revenue ($56,010) lags Spring ($60,840) by $4,830 — the widest seasonal gap.

› **Action:** Holiday campaigns (Nov–Jan) targeting Outerwear, Coats, Gloves; Spring new-arrival launches for Clothing.

› **Impact:** Closing 50% of the Winter gap alone adds ~$2,400 per season.

4. Launch a Tiered Loyalty Rewards Programme

› **Evidence:** 47% of customers are already "Loyal" (21+ purchases) — retain and formalise this relationship.

› **Action:** Introduce Silver / Gold / Platinum tiers with exclusive discounts, early access, and free shipping.

› **Impact:** Even modest increase in purchase frequency for 1,832 loyal customers drives significant revenue uplift.

5. Rationalise the Discount Strategy

› **Evidence:** 42.8% of transactions used discounts — unsustainably broad without a targeting framework.

› **Action:** Focus discounts on high-sensitivity seasonal items (Gloves, Coats, Shorts); add minimum-spend thresholds.

› **Impact:** Redirecting 20% of discount budget to higher-ROI segments reduces cost while maintaining revenue.

6. Premium Shipping Incentives for High-Spend Buyers

› **Evidence:** Express shipping users average $62.50 vs $58.90 for Free Shipping — a $3.60 spend premium.

› **Action:** Offer complimentary Express upgrade on orders above $80; bundle Free Shipping with $60+ minimum spend.

› **Impact:** Upselling shipping tier could increase average basket for ~30% of transactions.

8 CONCLUSION

This project demonstrates a complete, production-quality data analytics pipeline applied to a retail customer dataset of 3,900 records. Using **Python, PostgreSQL,** and **Power BI** in combination, the analysis surfaced actionable insights across every key dimension of the customer journey — from demographics and product preference to loyalty, discounting, and payment behavior.

Summary of Findings

› The dataset required minimal cleaning — only 37 missing values (Review Rating) were imputed using category-wise means, confirming high data quality.

› Male customers dominate by volume but equal per-capita spend across genders means female acquisition is a pure growth lever, not a risk.

› Subscription programmes are demonstrably effective — a 15% spend premium per subscriber justifies significant investment in programme growth.

› 47% loyal customers form a strong retention backbone, but only 16% are new, flagging the need for a more aggressive acquisition strategy.

› Discount penetration at 42.8% is high; surgical targeting toward high-sensitivity seasonal items would improve margin without reducing satisfaction.

› Seasonal and category patterns provide a clear roadmap for inventory planning and targeted campaign scheduling throughout the year.

Learning Outcomes

Skill Area	Applied Technique
Data Cleaning	Null imputation (groupby mean), column normalisation, duplicate removal
Feature Engineering	Age group binning (pd.qcut), frequency-to-days numeric mapping
SQL	Aggregations · Subqueries · CASE WHEN · CTEs · RANK() OVER PARTITION
Visualisation	Histograms · Bar/Donut charts · Horizontal bars in Matplotlib
Power BI	KPI cards · Clustered charts · Donut charts · Interactive slicers
Communication	Translating statistical patterns into business-oriented recommendations

Project Complete

This report was produced as a portfolio-quality data analytics project demonstrating Python EDA, SQL analysis, and Power BI dashboard development. All insights are derived exclusively from the provided customer dataset. Suitable for professional portfolio, LinkedIn showcase, and academic submission.

Analyst: Aditya • Tools: Python | PostgreSQL | Power BI • 2024